Jothi Learning White Paper Series

THE 4T TEACHING ENGINE

The Instructional Model of the

Jothi Tutor Academy

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Purpose of the Document

This document defines the 4T Teaching Engine, the instructional model used in lessons delivered within the Jothi Tutor Academy. It explains the logic of the model, the role played by each phase, and how the engine connects philosophy to classroom practice.

Intended Audience

This document is intended for:

- academy leadership
- mentors
- tutors delivering lessons within Jothi Learning
- internal readers seeking to understand the academy's instructional model

Related JTA Documents

- JF01 — Jothi Tutor Academy Executive Foundation
- JF02 — Jothi Trifecta Philosophy
- JTS01 — Lesson Delivery Standard
- JTD01 — Tutor Competency Framework
- JTD02 — Tutor Lesson Reflection Standard
- JTD04 — Mentor Observation Standard

1. Introduction

The 4T Teaching Engine is the instructional model used within the Jothi Tutor Academy. While the Jothi Trifecta explains the philosophy of learning, the 4T model explains how tutors teach in a way that makes student thinking visible and improvable.

The model prevents lessons from collapsing into one-way explanation. It gives tutors a disciplined cycle for introducing ideas, checking understanding, observing reasoning, and correcting misconceptions.

2. Overview of the 4T Teaching Engine

The four phases are:

- Teach
- Test
- Track
- Transform

Teach
↓
Test
↓
Track
↓
Transform

[Diagram placeholder — Insert 4T Teaching Engine diagram here]

These phases are not intended as a rigid one-pass checklist. They form a repeated instructional cycle that can recur multiple times within a lesson.

3. Teach — Concept Introduction

The Teach phase introduces a new idea, method, or piece of reasoning. The tutor explains the concept clearly, models the logic step by step, and makes the thinking visible.

Effective teaching in this phase includes:

- logical sequencing of explanation
- clear board work or written modelling
- worked examples that expose reasoning, not only answers
- language that highlights why each step makes sense

This phase is closely connected to the Strategy dimension of the Trifecta.

4. Test — Diagnostic Questioning

The Test phase checks whether the explanation has actually produced understanding. Tutors do not rely on vague confirmation questions such as “Do you understand?” They ask questions that reveal reasoning.

Examples include:

- “Why does this middle term become 2ab?”
- “What changes if this value is negative?”
- “Which step in this solution is most likely to go wrong, and why?”

Diagnostic questioning is essential because it turns hidden thinking into visible evidence.

5. Track — Monitoring Student Thinking

The Track phase involves observing how the student is applying the idea. The tutor watches written work, spoken reasoning, hesitation, and error patterns in order to understand what the student is actually thinking.

Tracking may happen through:

- board work
- written responses
- interactive platforms such as Classkick
- verbal explanation during problem solving

This phase matters because correct answers alone do not always show secure understanding.

6. Transform — Correcting Misconceptions

The Transform phase addresses the misunderstanding revealed during Test or Track. The tutor does not merely supply the answer. The tutor identifies the source of the error, clarifies the reasoning, and guides the student toward a more accurate understanding.

Effective transformation may involve:

- re-explaining the concept in a clearer way
- contrasting correct and incorrect reasoning
- using a better example

- asking the student to articulate the corrected idea

This is the phase where teaching becomes genuinely responsive.

7. The Engine as a Continuous Cycle

The 4T model should be understood as an engine rather than a sequence of isolated events. In a strong lesson, the tutor may move through the cycle several times as new ideas emerge or as different misconceptions appear.

This repeated cycle is what allows lessons to remain interactive, diagnostic, and concept-led.

8. Relationship to Tutor Development

The 4T Teaching Engine is not only a lesson model. It is also a training model for tutor development.

- JTS01 translates it into lesson-delivery expectations.
- JTD02 uses it as the structure for tutor reflection.
- JTD04 uses it as the structure for mentor observation.
- JTD01 defines competencies required to apply it consistently.

This is one reason the model functions as the core teaching spine of the academy.

9. Indicators of Effective 4T Teaching

A tutor is applying the 4T model effectively when the lesson shows the following features:

- explanations are clear and logically sequenced

- questions reveal student reasoning
- student thinking is actively monitored
- misconceptions are addressed promptly and clearly
- the tutor adjusts instruction in response to evidence

10. Why Diagnostic Questions Matter

Diagnostic questions deserve special emphasis because they connect teaching to evidence. Without them, a tutor may believe learning has occurred when the student is only following procedures mechanically.

Diagnostic questions help the tutor locate misunderstanding early and prevent weak reasoning from becoming embedded.

11. Relationship to the Wider JTA System

The 4T Teaching Engine sits at the centre of the academy's instructional model. It connects the philosophical foundation of the Trifecta to the practical expectations of teaching standards, reflection, observation, and competency development.

In this sense, the 4T model is not an isolated teaching tool. It is the operational spine that links lesson delivery to tutor development across the whole JTA system.

12. Conclusion

The 4T Teaching Engine is the instructional heart of the Jothi Tutor Academy. It gives tutors a disciplined method for moving from explanation into genuine understanding.

Its value lies in its responsiveness. It ensures that teaching is not only delivered, but checked, interpreted, and improved in real time.